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Deep learning–based intelligent system for railway track monitoring and fault detection

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Abstract

In a modern urban railway transport system, monitoring the track conditions without interruption is the guarantee of secure and reliable operation of the system. Inspectors who follow traditional methods generally lack the ability to make timely inspections at all times, are inconsistent in their evaluations, and often fail entirely to discover the most subtly incipient defects. In this paper, an approach to deep learning-based railway track monitoring and fault detection using an image classification technique is proposed. To address these deficiencies, this paper proposes a framework for monitoring railway freight manner tracks and detecting faults that uses deep learning through image classification approaches. The proposed method utilizes Convolutional Neural Networks (CNNs) to automatically extract discriminatory features from railway track images prepared under a variety of frequently changing operating and environmental conditions. Four pretrained CNN architectures, NASNetMobile, MobileNetV2, DenseNet121, and EfficientNetB0, are implemented and compared to find the most suitable model for fault detection. Experimental results show that EfficientNetB0 outperforms the other models with an accuracy 98.11%, a precision 97.04%, a recall 99.24%, and an F1-score 98.13%. The proposed method's reliability is further validated through thorough testing on unseen data. In addition, the optimized model is deployed using a Streamlit-based application so it can be interactively evaluated and visually demonstrated. This deployment illustrates the practical possibility of the suggested system for real-time railway track inspection, smart monitoring, and predictive maintenance applications.

Keywords Railway track monitoring, Fault detection, Deep learning, Convolutional neural networks, Image classification

1 Introduction

Railways are still the most commonly used and economically efficient type of transport on a global level and have an important role in economic development and social inclusion. From the point of view of railway systems' safety and reliability, one critical component sustaining this good level of performance is represented by railway tracks, whose components, rail lines in particular, are subjected to several degradation mechanisms such as cracks, misalignments, wear, or surface damages leading to derailment accidents, service disruption losses, and financial loss for infrastructure managers. Hence,



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the real-time and accurate monitoring of rails is in great demand for safe operation and maintenance [1].

Traditional railway track inspection relies on manual visual inspection (MVI) or specific inspection trains [2]. Even though these methods are frequently used, they are typically time-consuming, labor-intensive, and costly, and have a high risk of human error. Moreover, some faults cannot be inspected manually for real-time monitoring to detect an early-stage fault. As the scope and complexity of railway systems continue to grow, there is a potential need for intelligent, automated, and dependable low-maintenance monitoring solutions that can effectively operate in various environmental and operational contexts [3, 4]. It is interesting to note that in recent years, advances in artificial intelligence, especially deep learning techniques, have attained significant success when used for computer vision and image recognition. Convolutional Neural Networks (CNNs), which can learn hierarchical features automatically from images, avoid hand-crafted feature extraction and achieve high-precision fault detection [5]. These are the features that make deep learning a promising technology to apply to railway track monitoring: it can analyze visual data, is very accurate in discriminating between kinds of defects, and can do that at a constant level [6, 7].

This paper introduces A Deep Learning–Based Intelligent System for Railway Track Monitoring and Fault Detection, which relies on several CNN architectures to process the railway track images in order to identify fault status [8]. The performance of various deep learning models is compared using standard evaluation metrics, with the goal of identifying a successful one that can work effectively in real-time at practical systems. The method we present aims to increase rail safety, optimize maintenance, and decrease human interference in a fast and intelligent solution for automatic inspection of train tracks [9]. While the choice of pre-trained CNN-based architectures, namely EfficientNetB0 and MobileNetV2, has been reported in previous literature, the contribution of this work is not limited to just the selection of models but also towards a systematic multi-dataset, cross-component comparative evaluation approach. In contrast to prior works of model verification based on a single dataset, this work merges several publicly available railway datasets over varied track elements (rail surface, fasteners, and track structures), which allows for cross-domain generalization tests. Although there is increasing interest in deep learning-based railway inspection, how to identify an optimal CNN architecture that can perform reliable fault detection under different railway components and datasets still remains a question [10]. There are many studies that concentrate on a single dataset or model, which limits generalization and comparison. Furthermore, there is a requirement for an efficient and easy-to-use setup that allows for comparative analysis of the already pretrained models on the railway track fault detection problem [11].

With this aim, the presented framework pays attention to model generalization by testing performance on various datasets and railway components. Moreover, the work contains a comprehensive comparison since it covers not only accuracy metrics, but also model efficiency, convergence characteristics, and deployment practicality with other methods under consideration, which enrich the analytical evaluation. In order to overcome these issues, this paper presents a deep learning-based intelligent railway track monitoring and fault detection system based on CNN [12]. Four pre-trained CNN networks (NASNetMobile, MobileNetV2, DenseNet121, and EfficientNetB0) are applied,

and the performances of these models are compared in terms of their classification accuracy on defective and non-defective railway track images. The system is implemented and experimented with via a Streamlit application to build an interactive environment for image-based fault analysis. We have been focused on creating a flexible and efficient pipeline for automatic railway track inspection, identifying powerful deep learning models for deployment in the field. The paper also investigates the weaknesses of the system, such as low image quality sensitivity, dataset diversity, and environmental viability, as well as scalability and practical deployment. This reflexive consideration gives the analysis added depth and practical salience.

1.1 The key contributions of the proposed model are:

- Public railway inspection data on rail surfaces, fasteners, and track components is used to perform a comprehensive evaluation of the proposed method for fault detection across various types of defects as well as operating circumstances.
- The creation of an intelligent railway track fault detection method combining deep learning, which has a feature of computational efficiency while preserving high classification accuracy in image-based analysis through the EfficientNetB0 model.
- Transfer learning is done by using pre-trained ImageNet EfficientNetB0 weights to enhance feature extraction and reduce the training complexity for the railway tracks datasets.
- The developed model is being deployed in a Streamlit-based application, and a user-interactive interface has been implemented for uploading and analyzing the railway track images.

2 Related works

Railway infrastructure inspection is a subject that has received extensive research attention because the integrity of the railway is critical to its safety and reliability. The traditional manual detection approaches are relatively slow, expensive, and susceptible to human errors, raising automation needs. Recent advances in the field of computer vision and deep learning have empowered an efficient visual comprehension of railway components. Recently, convolutional neural networks have been widely used in defect detection of track images. These approaches show better precision and effectiveness compared with existing inspection methods. Nevertheless, most of the current studies are still restricted to small datasets, the lack of cross-domain validation, and few discussions on model generalization and deployment challenges. In this work, we fill this gap by comparing a number of deep learning architectures over diverse railway datasets and discussing their tradeoff between generalization abilities and computational efficiency.

The study in [13] designed a smart railway traffic safety system using object detection, segmentation, machine learning, and automatic notification. Railway boundaries detection and intruder identification across different environments were achieved using the Mask R-CNN and YOLOv3, while XGBoost performed with high precision to predict the size of an intruder. The model performs well in controlled environments and can alert timeously via a message service. Nevertheless, the method is constrained by the requirement of fixed camera viewpoints, high computational cost, and a lack of successful real-world deployment and long-term evaluation.

In [14], the authors propose a machine learning technique to detect railway track faults by using vibration data collected from in-service vehicles. The review classifies data sources, preprocessing methods, fault prediction approaches, and model structures employed in previous studies. Although a small amount of real-world data enhances reliability, the challenges, such as insufficient public datasets availability, complex data preprocessing, and label validation difficulties, still remain. Some of the main findings are that for deep learning methods, they are too slow and costly to be used with current GPUs, a lack of possibilities for real-time deployment, as well as a lack of exploration on exploiting these techniques in linear time and on unsupervised multifault detection.

The work [15] deals with the safety of rail by presenting an integrated AI and IoT-based power system for the detection of railway track cracks and automatic gate control. An infrared transmitting-receiving pair and a servo motor are used to automate the gate opening operation according to the movement of the train. A crack detection DNN using Residual DenseNet is developed to recognize cracks from the sample visual inputs on cloud servers. The identified cracks are sent in real time via GPS and GSM modules, while visualization of the data is ensured by Firebase Cloud for improved safety of the railway and accident reduction.

In [16], approaches via image processing to detect defects in rolling stocks and tracks of the railway system for safe train operation are investigated. It shows the drawbacks of manual inspection and demonstrates the benefits of machine vision-based inspection systems. The review discusses the application of computer vision, machine learning, and deep learning algorithms, and image acquisition and preprocessing techniques for improving the accuracy of detection. It also points out the research gaps, such as the establishment of standard datasets and benchmark methods, and future work of improving railway defect detection systems [17].

Researcher in [18] overcomes the drawbacks of conventional track fault detection of railways by using artificial intelligence (AI)-based image classification methods. A new CTGAN-based image augmentation technique is presented, which can generate photo-real railway track images and increase the diversity of data. Several deep learning models are constructed and compared, utilising hyperparameter search and k-fold cross-validation for evaluation consistency. The computational expense of training generative and deep learning models restricts their real-time deployment on resource-limited platforms.

A prescriptive analytics model for intelligent predictive maintenance of railway tracks with machine learning and graph convolutional networks is presented in [19]. The system decodes geometric parameters and maintenance history in order to forecast possible defects before the track failure happens. The method, which allows for proactive maintenance scheduling and is designed to minimise downtime / maximise the lifetime of the rail network infrastructure. The effectiveness of progressive machine learning algorithms to support railway maintenance efficiency and safety is demonstrated experimentally.

The authors of [20] consider predictive health maintenance of rail infrastructure, applying nondestructive testing and imaging. Pre-trained models with transfer learning are used to overcome the problem of insufficient datasets in fastener inspection. Various machine learning models, such as ViT, DeiT, VGG16, VGG19, and ResNet50, are compared in terms of fault identification. Transformer and Transformer-Based Models have

better generalization, whereas CNN-based models provide us with accuracy–computation trade-offs.

In [21], an AI and IoT-based railway track fracture detection system is proposed using a CNN for image processing. The product system includes a camera, IoT communication, and alert functions that inform maintenance of the operating state, through a cloud platform and field indicators. This method eliminates visual crack detection and enhances the efficiency of inspection activity, safety of airframe structure, and the maintenance turnaround time. But the system is constrained by image quality, environmental factors, network issues, and, critically for real-life use, large quantities of well-labelled data.

A neural network whose function is to detect the railway track using CNNs and image processing techniques for automation of detection in monitoring track's abnormality. It monitors visual information in real time for the purposes of the track component identification, damage detection, and obstacle discovery [22]. It also reduces human-based errors, the time to respond, and related operating costs by removing the need for manual field patrols. But its performance is highly dependent on the diversity of training data and scene view-points as well as ambient conditions, and -depending- its use in real-time may come with a CPU-GPU server/comms cost [23].

The study in [24] presents an automatic monitoring system for the braking system of a train wagon based on object detection models YOLOv1. The You Only-Look-Once (YOLO) is a real-time object detection model for increasing railway safety. Several YOLOv11 variants are trained for classifying the brake condition in order to mitigate risks such as wheel damage, fires, and accidents due to brake failures. The experimental findings indicate that different YOLOv11 slices are preferable based on whether the false negatives or the false positives are of concern. Nevertheless, the method has limitations on dataset size, class imbalance, and the computational cost of the larger YOLOv11 model for real-time applications.

In recent years, a variety of AI- and IoT-based solutions have been proposed for railway safety, such as vision-based intrusion detection, vibration analysis, crack detection, activity monitoring for predictive maintenance, and braking system monitoring. Deep learning algorithm-based models, like CNNs, transformers, GCNs, and YOLO variants, have shown better accuracy and automation than the classical inspection methods. Data-driven/machine learning solutions enable early fault detection, predict maintenance, and improve safety. On the other hand, most of these methods are in need of control environments, such as predefined sensor or camera settings, and datasets that fail to cover all possible situations used for training. The computation overhead is high, the real-time large-scale deployment is limited, and it still needs a large amount of high-quality labeled data for the training process. These disadvantages make it evident that a lightweight, scalable, and general intelligent monitoring system should be implemented, considering different rail real-world environment cases.

3 Proposed methodology

In this paper, we propose a deep learning-based automatic railway track defect detection system. The proposed approach combines imaging, feature extraction, and classification into a useful algorithm to distinguish defect-nondetect states of tracks. A small convolutional neural network is designed to reduce computation and increase accuracy

in fault detection. One of the key aspects for safe and reliable rail traffic is railway track condition assessment, where visual inspection plays a suitable role as part of manual or automated monitoring systems. Figure 1 shows a visual observation of a defective and non-defective railway track condition. The top row of panels labeled “Defective Image” illustrates representative damaged tracks, in which visible rail cracks and fractures, a missing rail wearing plate on the concrete sleeper, and misalignment between the rail and sleepers due to broken elastic fastener/rail clipping are shown. These problems are signs of structural defects, which can lead to instability of the track and its safety. In this regard, it is noted that the bottom row in Fig. 1 shows “Non-Defective Images” illustrates railway lines where the rails are continuous and intact with good alignment supported by sound sleepers properly packed on ballast to be able to sustain its traffic.

The dataset consists of 2,968 images containing an equal number of defective and non-defective samples, 1484. Among these images, 2,140 are for training and 564 and 264 for the validation and test sets, respectively. The experimental dataset is constructed by integrating Dataset 1_Rail [26], Dataset 2_Fastener [27], and Railway Track Fault Detection [28] into a more variable railway inspection dataset. This composite dataset contains images of rail surfaces, fasteners, and different track components with defective and non-defective conditions under various real-world appearances. The aggregation of several datasets makes the data diversity richer, the model generalization better, and facilitates robust evaluation on railway track fault detection performance in heterogeneous tracks. The class-balanced distribution is preserved in all the subsets and contributes to reducing learning bias. The validation set is used for tuning the model, and the test set, not seen in training, serves for the final evaluation of performance, indicating a robust evaluation of the approach. The images are resized to a fixed resolution before training and normalized to stabilize the learning. The approaches also include data augmentation that involves rotating, flipping, and varying the brightness of input images in order to enhance model generalizability [25].

The architecture of the EfficientNetB0-based railway track defect detection model is shown in Fig. 2. Starting from the input railway track images, the model first processes them with a 3×3 convolution layer to obtain primary low-level features, including edges and textures. This is followed by a stack of Mobile Inverted Bottleneck (MBCnv) blocks,



Fig. 1 Sample defective and non-defective rail track

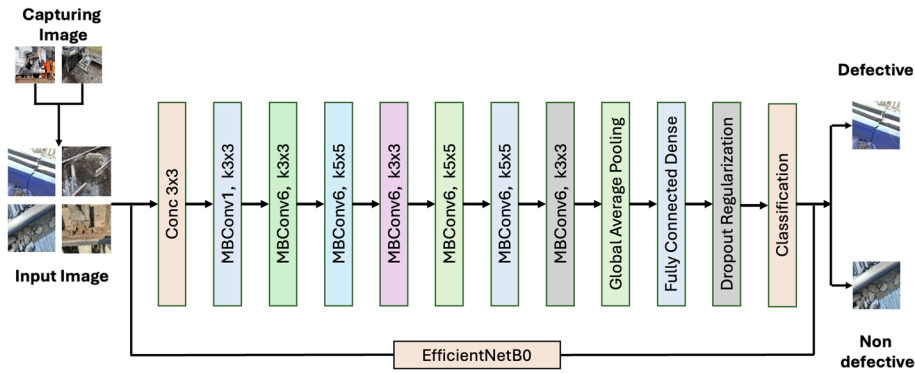


Fig. 2 EfficientNetB0-based railway track defect detection model

which start with MBConv1 3×3 and then several MBConv6 with respective 3×3 , and/or 5×5 kernels. MBConv blocks are used in combination with depthwise separable convolutions and squeeze-and-excitation operations to learn fine-grained as well as coarse-grained information from the track images simultaneously. 3×3 kernels are useful to find small defects, such as the crack in a leaf itself, and 5×5 kernels allow for a larger receptive field, which is useful to detect structural faults, like breaks and misalignments. Following the feature extraction, global average pooling is used to downsample the spatial dimensions and reduce parameters. The resultant feature vector is then propagated through a fully connected layer and dropout regularization to avoid overfitting. Ultimately, the output is passed to a classification layer, which classifies the railway lines into defective and non-defective classes. The following are steps on how the EfficientNetB0 model works for railway track detection.

3.1 Input layer

First, we resized every image in a railway track video sample to 224×224 pixels. The downsampled image is expressed as a 3-D tensor in Eq. (1).

$$X \in R^{224 \times 224 \times 3} \quad (1)$$

Here, 224×224 is the height and width of the image. The value 3 represents the RGB color channels of the input to our model.

3.2 Pre-trained EfficientNetB0 feature extractor

The input images are passed through a pre-trained EfficientNetB0 model as a feature extractor. EfficientNetB0 is a small convolutional neural network trained on the ImageNet dataset, and learned visual patterns such as edges or shapes. The classification head is discarded, and we keep only the convolutional backbone. The feature extraction can be formulated as in Eq. (2).

$$F = f_{EffNet}(X) \quad (2)$$

where f_{EffNet} denotes the composition of convolutional, batch normalization, and activation layers, and F is the high-level feature map. The parameters of the EfficientNetB0 model were fixed during training to maintain learned representations and save computational cost, formulated as in Eq. (3).

$$\frac{\partial L}{\partial \theta_{EffNet}} = 0 \quad (3)$$

This permits the model to make use of ImageNet features and avoid overfitting, particularly with limited training data, and also speed up training. The resulting feature maps F are subsequently fed to task-specific layers for the ultimate classification or decision.

3.3 Global average pooling

After sophisticated feature extraction, we adopted the Global Average Pooling (GAP) operation to map the complex spatial feature maps into a concise and discriminative feature vector. The resulting feature map from the convolutional backbone -denoted as $F \in R^{H \times W \times C}$ where (H), (W) represent the spatial dimension of a feature map, and (C) is the number of channels in the particular layer. Global Average Pooling works by finding the mean value of every feature channel across all spatial locations. It can be formulated mathematically in Eq. (4).

$$g_c = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W F(i, j, c) \quad (4)$$

For each channel $c \in \{1, 2, \dots, C\}$, as a result, this operation compresses the two-dimensional spatial information into a single scalar value for each channel to generate a feature vector shown in Eq. (5).

$$g \in R^C \quad (5)$$

This operation directly captures the occurrence and contribution of each learned feature across the entire image, so that we obtain a representation invariant to small spatial shifts and deformations. Unlike traditional fully connected layers, the Global Average Pooling significantly reduces the overall number of trainable parameters and consequently decreases the computation overhead and memory cost by an order of magnitude. Moreover, as one of the dimensions (patch size) becomes infinite after performing GAP, no overfitting and specialization to small patches can occur. These unique characteristics illustrate the high appropriateness of Global Average Pooling for classification applications, particularly in problems where powerful high-level feature representations are crucial.

3.4 Fully connected dense layer

The aggregated feature vectors are then fed to a fully connected dense layer with 128 neurons for learning task-dependent representation. This is defined as in Eqs. (6) and (7).

$$z = W_1 g + b_1 \quad (6)$$

$$a = \text{ReLU}(z) = \max(0, z) \quad (7)$$

where W_1 and b_1 are trainable weights and bias. The non-linearity introduced by ReLU activation helps the model to learn complex patterns and concentrate on significant features for railway track defect detection.

3.5 Dropout regularization

The dropout regularization technique is used to reduce overfitting by randomly disabling 40% the neurons in the dense layer during training, which can be expressed as in Eq. (8).

$$a' = m \odot a, m \sim \text{Bernoulli}(0.6) \quad (8)$$

In this case, m is a binary mask, and each neuron is kept with probability=0.6. This mechanism leads to strong and invariant feature learning by discouraging dependence on specific neurons.

3.6 Output layer

For this, the classification uses a single sigmoid-activated neuron that returns a probability for the railway track image. The output is calculated as in Eqs. (9) and (10).

$$y = \sigma(W_{2a} + b_2) \quad (9)$$

$$\sigma(x) = \frac{1}{1 + e^{-x}} \quad (10)$$

Here W_{2a} and b_2 are the learnable weights and biases. Since the sigmoid function produces results ranging between 0 and 1, if $y = 1$ refers to a defect track, while $y = 0$ it signifies a non-defect track. This output produces decisions with which to make confident defect detections.

3.7 Loss function

Loss function used the Binary Cross-Entropy (BCE) loss function to calculate the difference between the true label (y) and the predicted probability $\hat{y}$ shown in Eq. (11).

$$L = -[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})] \quad (11)$$

This loss is particularly punishing for the model as regards confident, yet erroneous predictions, and makes it more reliable when deciding between inputs corresponding to non-defective or defective railway track images.

In the model training stage, the binary cross-entropy loss function is minimized through the Adam optimization method to adjust parameters of the dense and output layers, while using the EfficientNetB0 backbone as a frozen way for retaining pre-trained feature representations. Training is performed over multiple epochs with mini-batch gradient updates to iteratively improve prediction. Dropout regularization and early validation monitoring are used to avoid over-fitting and guarantee stable convergence and better generalization. In order to be reproducible and fairly comparable, all models are trained in the same experimental conditions with the same unified dataset. We utilize transfer learning with ImageNet seed weights to increase feature extraction efficiency and training ease. The Adam optimizer uses the learning rate of 0.0005, batch size of 32, and is trained for only up to 20 epochs. Binary cross-entropy is employed as the loss function, and early stopping is performed to prevent overfitting. Moreover, the data augmentation strategies, including horizontally flipping, rotating, zooming, and contrast adjustment, are used to enhance the robustness of our model and achieve good generalization.

4 Results and analysis

This section carefully describes our empirical results, showing the details of each experimental result achieved using a novel deep learning model tailored to the complex problem of railway track defect classification. The former dataset was carefully divided into three separate subsets, including the training set, validation set, and testing set, which are approximately 72%, 19%, and 9% of the entire dataset, respectively. The model is extensively trained with the training data, so that it can learn well to discriminate features from various types of images of rail tracks. The validation data are selectively used throughout the training process to supervise the model's performance and fine-tune important parameters, which is just-in-time. Then the ultimate test is made on a real-world independent dataset, which plays an important role in showing how good our model is in the real world. The performance of the model is critically examined using accuracy, precision, recall, and F1 score as different performance measures. The proposed method is very effective for classifying defective and non-defective conditions of tracks, which clearly indicates its feasibility in real-world maintenance/inspection aiming at railway system safety assurance.

Figure 3 shows how NASNetMobile, MobileNetV2, DenseNet121, and EfficientNetB0 perform while learning over training epochs using training and validation accuracy. On the train accuracy graph, as can be observed from the figure, all models steadily improve, suggesting that feature learning on railway track images has been successful. EfficientNetB0 also maintains the highest training accuracy, almost at near perfection, and indicates more powerful feature extraction. DenseNet121 and MobileNetV2 exhibit a relatively modest increase, reaching a similar optimal performance plateau at a somewhat lower accuracy rate, although NASNetMobile learns more slowly and converges to the largest gap of training accuracy. In the validation accuracy plot, EfficientNetB0 has always been holding the best validation accuracy with a stable convergence, meaning good generalization and no over-fitting. DenseNet121 and MobileNetV2 have a modest validation accuracy with small oscillations, indicating that they have a fair generalization performance, whereas NASNetMobile has the lowest validation accuracy, signposting weaker performance. These results confirm that EfficientNetB0 outperforms other networks both in training and validation processes, proving the robustness and efficiency of this network for railway track fault detection.

Comparison of loss convergence. Figure 4 illustrates the comparison of training and validation reduction rate against iterations, in NASNetMobile, MobileNetV2, DenseNet121, and EfficientNetB0 architectures. The left one depicts the training loss curve in which all models get a decreasing curving for learning and optimization. The

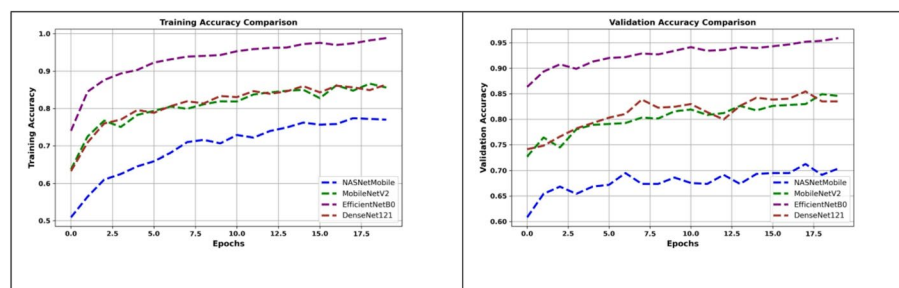


Fig. 3 Comparison of training and validation accuracy for railway track fault detection deep learning models

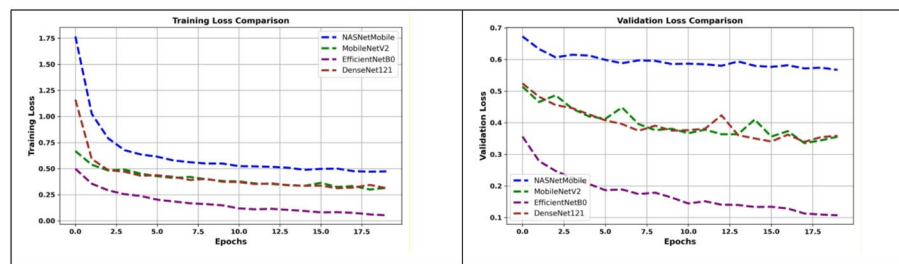


Fig. 4 Comparison of loss trained and validation for railway track fault detection using deep learning models

Fastest learning, the best to extract feature, and convergence. Faster Loss reduction. EfficientNetB0 shows the fastest and most stable decrease in training loss, achieving the lowest loss, which indicates better feature learning and convergence. DenseNet121 and MobileNetV2 exhibit medium general loss reduction with steady convergence to higher values, as EfficientNetB0 does, while NASNetMobile holds the highest training loss at all time frames -indicating slower learning. The right graph shows validation loss, which presents generalization on data not seen before during training. In particular, we observe that EfficientNetB0 achieves the lowest and most stable validation loss, hence generalizing well to the unseen data with less overfitting. DenseNet121 and MobileNetV2 have weaker stability with mild fluctuation in validation loss, making the generalization performance mediocre among all types, but NASNetMobile has the worst generalization ability. This figure indicates that the EfficientNetB0 can have better optimization performance the lowest loss of training and validation, which verifies its effectiveness and robustness in accurate detection of railway track fault.

The confusion matrices of the four deep learning models applied in railway track defect classification are shown in Fig. 5, and their decision behaviour is characterized with respect to defective and non-defective samples. Each matrix provides a view on which the predictions are distributed over truth and falsehood, going beyond overall accuracy. The finding is that it is difficult to separate subtle defect patterns, so the NASNetMobile has a higher misclassification. MobileNetV2 achieves better discrimination, but there is still clear confusion between categories. DenseNet121 performs better in terms of balance, indicating that there are relatively fewer false predictions and there is a more stable recognition of defects. EfficientNetB0 obtains the most favorable distribution with a large mass concentrated near the diagonal, indicating reliable classification. Fewer false positives indicate better resistance against normal track variances. Testing have a lower number of false negatives, then the more sensitive to physical defects your exam becomes, which is good for safety. Inter-model differences arise from architectural ability to represent fine-grained visual information. In general, the results in this work highlight that EfficientNetB0 is the most stable and reliable for the classification of railway track inspection.

Figure 6 Receiver Operating Characteristic (ROC) curves of four deep learning models are provided in the plot for railway track defect classification, showing their discriminatory performance at different decision thresholds. Each ROC curve demonstrates the trade-off between true positive and false positive rates, with a view to classification robustness across threshold settings. NASNetMobile shows a gradient lip having a lower curve elevation, which results in less separability between defective and non-defective classes. The high quality of the classification is reflected by the better

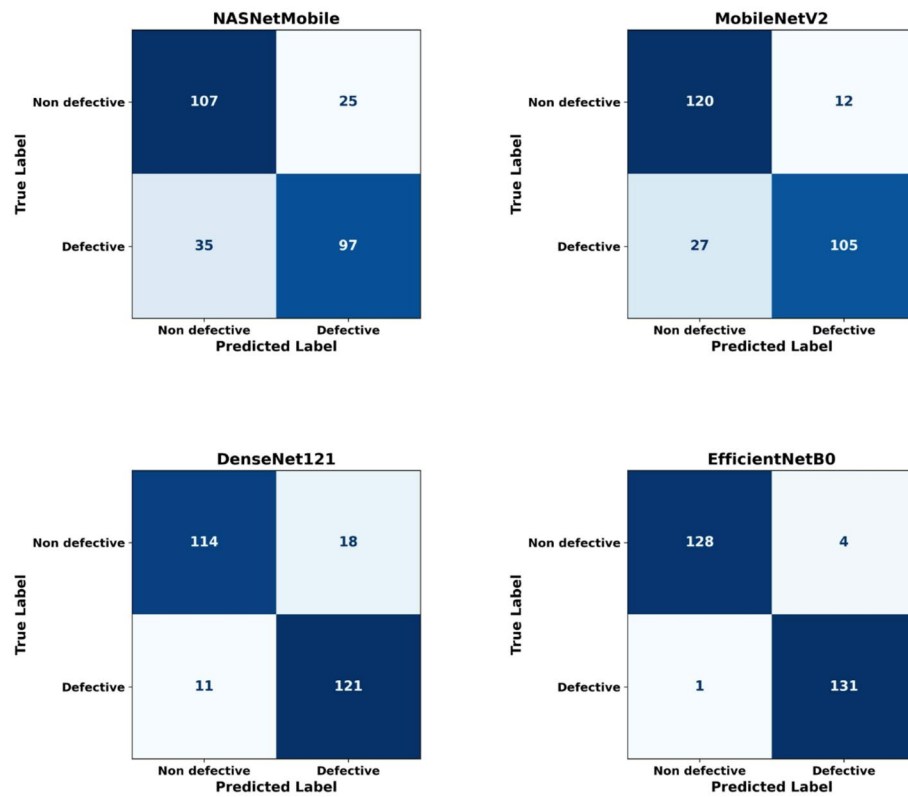


Fig. 5 Confusion matrices of deep learning models for railway track defect classification

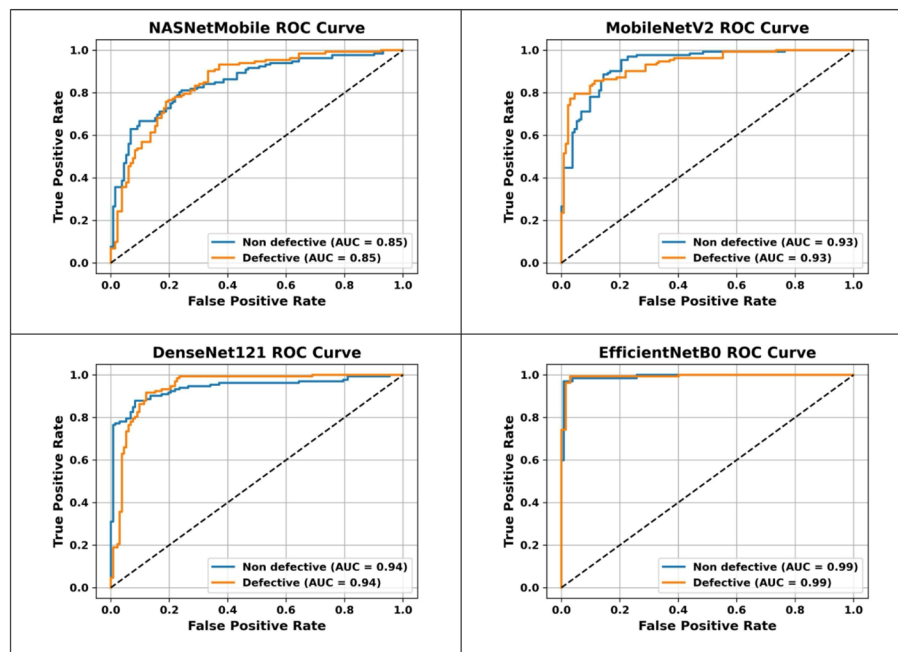


Fig. 6 The ROC curves of different deeplearning models for railway track defect classification

discrimination between classes, which can be explained by an improved learning of features. DenseNet121 reinforces this pattern again with higher AUC values and stable prediction behavior. The ROC curve of EfficientNetB0 comes to the top-left corner as close as possible. The aforementioned high AUC of EfficientNetB0 indicates its consistency in differentiating defect conditions. Identical AUC values demonstrate balanced sensitivity to defect and non-defect samples. The comparison also shows the effect of architectural efficiency and feature reuse on classification performance. The ROC analysis confirms that EfficientNetB0 is reasonably reliable for practical railway track fault detection tasks.

Figure 7 presents the precision–recall (PR) curves of four different deep models applied for railway track defect classification and illustrates their performance via sub-class sensitive testing. The precision–recall analysis provides a reliable method to evaluate the reliability of defect detection, especially when its misclassification cost is asymmetric. Notably, NASNetMobile has worse performance under high recall levels and lacks robustness when detecting anomalous samples. MobileNetV2 manifests better trade-offs, with a higher precision at most recall values. DenseNet121 achieves the best performance, which can be attributed to its stronger feature representation and more accurate defect recognition. EfficientNetB0 has the best precision–recall curve aligned along the upper bound for a majority of recall values. High AUC of PR curve further demonstrates its robustness in recognition of the defective track state with fewer false alarms. The similar and nearly tied class-wise curves indicate the well-balanced learning over defective samples and non-defective samples. We can compare trends and see how the efficiency of architecture affects classification confidence. The PR curve analysis further validates that EfficientNetB0 is the most trustworthy model for practical railway track fault detection.

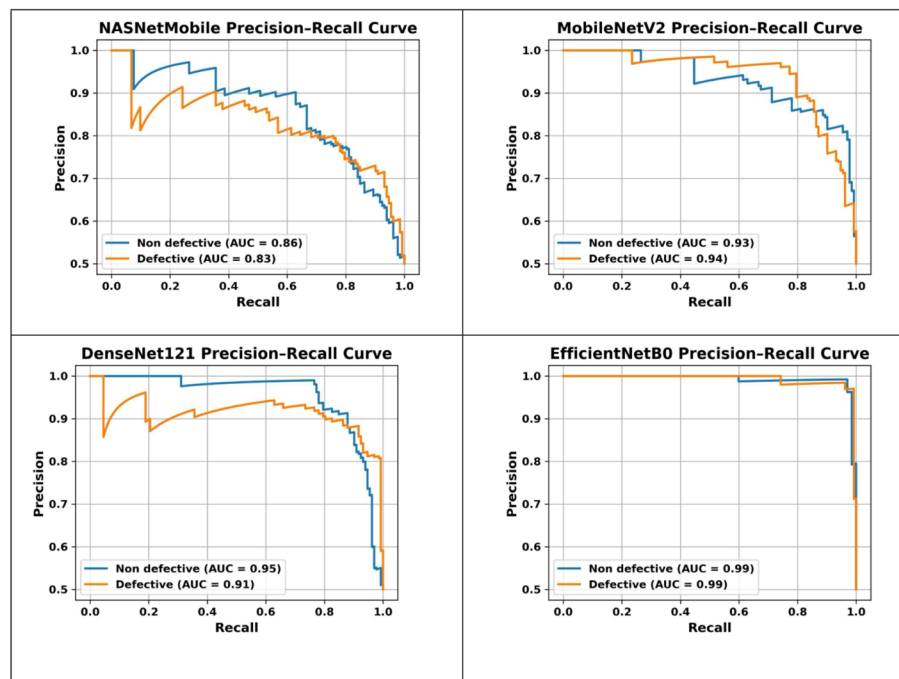


Fig. 7 The PR curves of different deeplearning models for railway track defect classification

Figure 8 shows the comparison of four deep learning models in terms of classification accuracy results for railway track fault detection. We find evident performance differences among various architectures, which demonstrates the impact of model design on detection capacity. The lowest accuracy is achieved by NASNetMobile, which achieves only 77.27% performance and shows the minimum ability of distinguishing defective and non-defective track state. MobileNetV2 has better accuracy, which is 85.23%, but it indicates poor feature learning capability, where avenues are still possible for further improvements. DenseNet121 further improves the classification efficiency, reaching 89.02% accuracy, due to superior feature reuse and learning stability. All the tested models, EfficientNetB0 achieves the highest accuracy of 98.11%, clearly surpassing others. This result demonstrates the strong feature representation learning and generalization ability on unseen test data. The incremental improvement in the models' development proves that architectural efficiency is key for railway track inspection. Overall, EfficientNetB0 stands out as the most accurate and consistent model for automated railway track fault detection.

To assess the recall performance, Fig. 9 provides a comparison of four deep learning models used in railway track fault detection. Our results show that there is significant variance in the accuracy of different architectures at localizing faulty track instances. NASNetMobile achieves the minimum recall of 73.48%, indicating that it has a greater chance to miss defect cases, which makes it less applicable for safety-critical applications. When comparing with MobileNetV2, achieves 79.55%, the latter is much greater than the former and also shows an outstanding improving tendency. With an average recall of 91.67%, DenseNet121 further enhances detection performance, and more faults are detected consistently. Comparing all the models, EfficientNetB0 achieves the highest recall 99.24%, indicating that it performs the best in reducing false negatives. This high recall performance proves that EfficientNetB0 works well for identifying most of the defective samples. The steady improvement over models highlights the importance of architectural design to reliable fault detection. Overall, EfficientNetB0 stands as the most reliable model for railway track defect detection in the recall dimension.

The Fig. 10 also shows a comparison of precision for four deep learning models employed in railway track fault detection. The precision indicates how reliable the

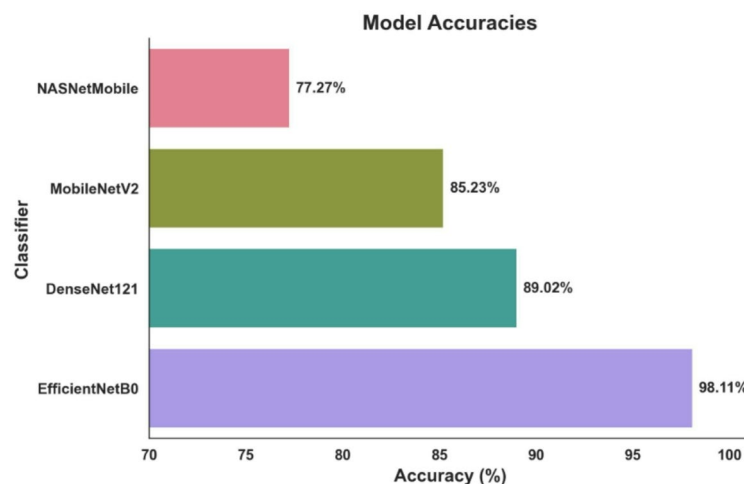


Fig. 8 Accuracy comparison of deep learning models for railway track fault detection

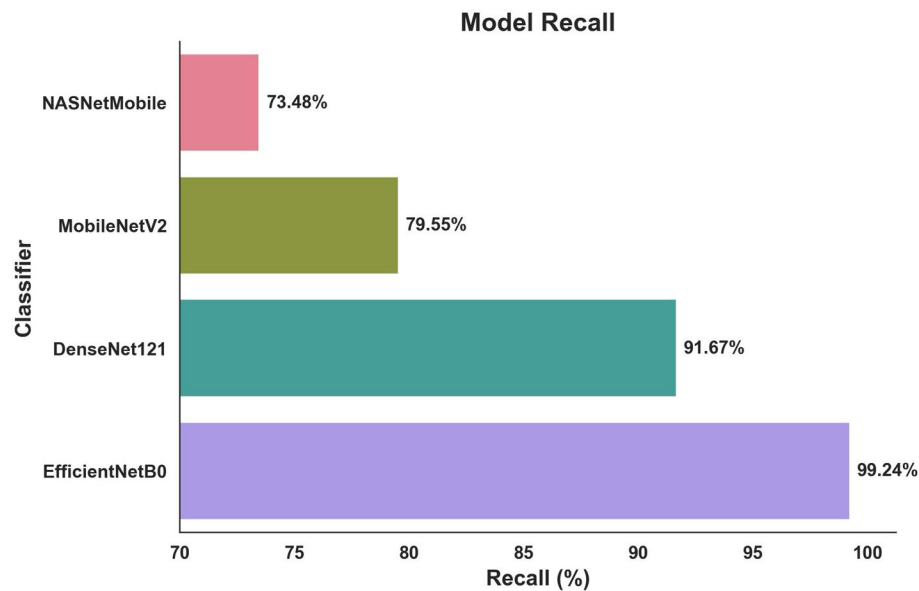


Fig. 9 Recall comparison of deep learning models for railway track fault detection

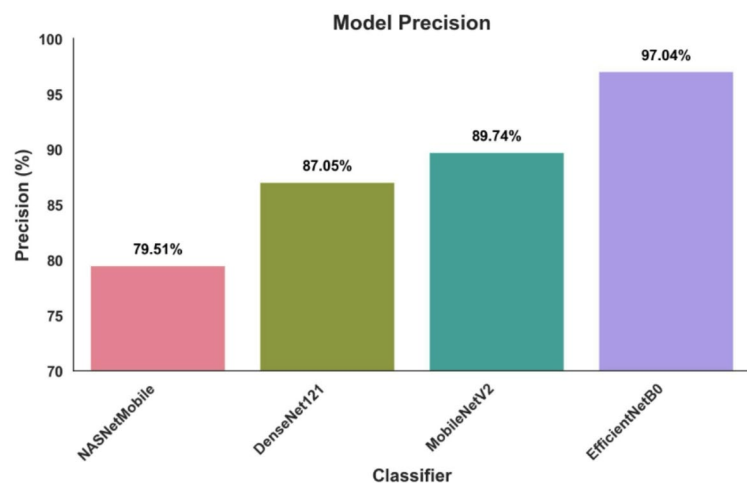


Fig. 10 Precision comparison of deep learning models for railway track fault detection

models are in recognizing the defective tracks without unreasonable false alarms. In particular, NASNetMobile shows the worst precision of 79.51%, implying more incorrect predictions on defect identification. DenseNet121 further gets a significantly higher precision 87.05% due to its strong discrimination ability and less false positive rate. For MobileNetV2, there is a marginal gain with a precision of 89.74%, implying better but still confined classification trustworthiness. EfficientNetB0 performs best among the other models, with a top precision of 97.04%. This provides better accuracy, which shows its strength in the defective and non-defective separation of track conditions. The results indicate that the best and most reliable performance in the railway track fault detection is obtained using EfficientNetB0, among the ones that have been tested.

Figure 11 compares the F1-score performance of 4 deep learning models on railway track fault detection. F1-score takes precision and recall into account, which gives a general measure of the classification performance. NASNetMobile has the lowest F1-score

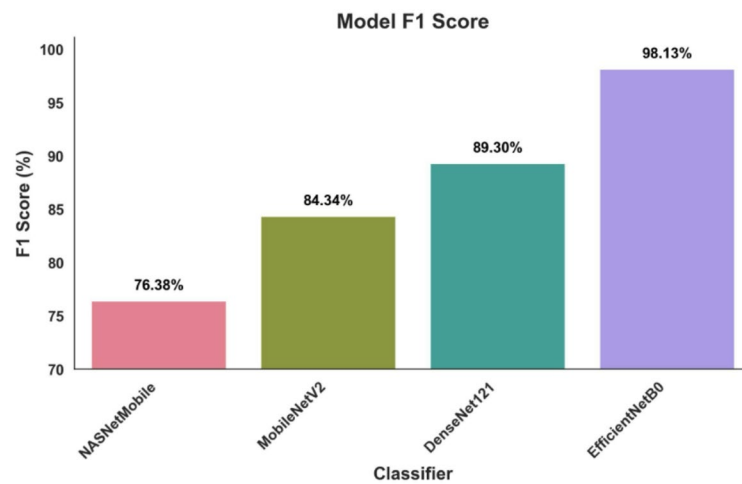


Fig. 11 F1 Score comparison of deep learning models for railway track fault detection

Table 1 Comparison of performance and inference time of deep learning models for railway track fault detection

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	Inference time (s)
NASNetMobile	77.27	79.51	73.48	76.38	10.3955
MobileNetV2	85.23	89.74	79.55	84.34	8.3339
DenseNet121	89.02	87.05	91.67	89.30	13.3867
EfficientNetB0 (Proposed)	98.11	97.04	99.24	98.13	7.6831

of 76.38%, meaning that there is not enough overall reliability in identifying faults. When using MobileNetV2, the results are more promising with an F1-score of 84.34%, showing that this model's performance tends to better balance the detection of defects against false predictions. In addition, DenseNet121 obtains better performance with an F1-score of 89.30%, which demonstrates that its feature representation is stronger. An optimal F1-score of 98.13% is achieved by EfficientNetB0, which is considerably better than the other models. This indicates that it strikes the best balance of sensitivity and classifier confidence. In general, the results reconfirm that EfficientNetB0 is the best-suited pre-trained neural network for fault detection on railway tracks across the introduced models.

The performance and efficiency comparisons of four deep learning models are shown in Table 1. The proposed EfficientNetB0 obtains the best overall performance, which delivers the best accuracy, precision, recall, and F1-score as well as consumes the least inference time, 7.68 s on the test set. This indicates that the EfficientNetB0 not only achieves better detection performance, but also gets better computational efficiency than NASNetMobile, MobileNetV2, and DenseNet121, and it is more feasible for real-time railway track monitoring.

Figure 12 shows the Rail Track Fault Detection System implemented by Streamlit as a local web-based application. Railway track images can be uploaded to the Streamlit interface by either dragging and dropping files into the window or by selecting desired files via the browse option. After you upload an image, the "Predict" button causes the trained deep learning model to predict the image.

In the left panel, an uploaded image is processed by the system, and it predicts a "Defective" railway track. The right-side different input images were classified as

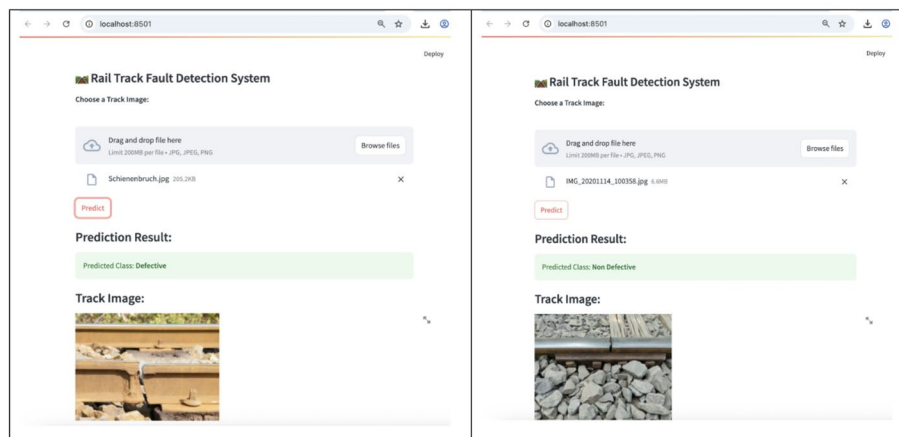


Fig. 12 Testing sample images with the web interface using the EfficientNetB0 model

“Non-Defective”, indicating the correct detection of a soundtrack condition. This visual comparison demonstrates that our approach is effective even in the presence of real-world inputs. In general, the Streamlit implementation shows that the railway track fault detection model we propose can be deployed practically and makes real-time inference and a user-friendly interface, suitable for aiding inspection and maintenance work.

5 Conclusion

This work presented a deep learning–based intelligent system for automatic railway track monitoring and fault diagnosis, to enhance the accuracy of inspections and safety performance. Several pretrained CNN architectures were applied and compared to find the optimal model for defect categorization. Comparative analysis of accuracy, precision, recall, and F1 scores showed obvious discrimination between the presented CNN model. Experiments revealed that the EfficientNetB0 always performed better than NASNet-Mobile, MobileNetV2, and DenseNet121 under all metrics of evaluation. EfficientNetB0 demonstrates high precision and well-balanced performance, which confirms that it has a good ability to learn discriminative features from rail track images. Its high recall points out weak false-negative detection, whereas high precision dampens false positives. These features are very necessary in the Safety–Critical Railway Inspection Systems. The robustness of the method indicates good generalization to other types of operational conditions. In addition, we make our trained model deployable, which can be accessed through a web application based on Streamlit for practical, real-time, user-friendly fault detection. This study illustrates how deep learning solutions can be integrated into modern railway maintenance. In general, the system proposed is desirable as an automated and smart replacement of former manual inspection approaches. Results confirm the potential of deep learning to improve the monitoring of railway infrastructure. Future work may include multi-class defect classification, video-based inspection, and applying the model in an edge device for field application with real-time performance.

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Author contributions

Study conception and design: Sujatha Krishna, Rajesh Natarajan; data collection: Rajesh Natarajan; analysis and interpretation of results: Francesco Flammini, Rajesh Natarajan; draft manuscript preparation: Sujatha Krishna, Rajesh Natarajan. All authors reviewed the results and approved the final version of the manuscript.

Data availability

The datasets used and/or analysed during the current study are available in the article.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent to publish

Not applicable.

Competing interests

The authors declare that they have no competing interests.

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